Adaptation-Aware Model Selection for Few-Shot Digital Twin Instantiation

This supplementary material provides additional implementation, analytical, reproducibility, and diagnostic details for the main manuscript, including counting conventions, candidate-bank and calibration records, analytical derivations, baseline configurations, hardware profiling, real-workload evaluations, and synthetic-benchmark generation settings.

The source, architecture-preselection, margin-calibration, diagnostic, and held-out target pools use disjoint center identities.

S1 Detailed Operation-Count and Parameter-Count Rules

The estimated operation count is computed for one forward pass with batch size one. Additions and multiplications in the main linear, convolutional, recurrent-matrix, residual-projection, and output-layer operations are counted. Bias additions, nonlinear activations, dropout operations, and GRU elementwise operations are excluded. The same deterministic rule is applied to every candidate.

For an instantiated architecture a_q, the parameter count is the exact number of elements in all trainable parameter tensors after the target input dimension and prediction horizon are fixed. Both counts depend on the target case only through its instantiated input and output dimensions. Because parameter adaptation does not change the architecture, both quantities remain unchanged.

The two indicators are constrained by hard deployment limits. They are not added to the prediction loss and are not changed after target adaptation. They are deterministic model-level indicators rather than direct guarantees of hardware-specific latency or memory use. Their relation to measured inference latency and serialized model size is evaluated in Supplementary Section S5.

S2 Candidate-Bank Records and Relative-Margin Calibration

S2.1 Candidate-bank construction and source-trained records

The initial architecture space contains 66 indexed configurations: 18 MLPs, 36 TCNs, and 12 GRUs. Configuration A57, a one-layer GRU with 32 hidden units, is fixed as the reference architecture before preselection.

An alternative configuration is retained when it records at least two check-oracle wins over A57, or at least two validation-selected positive check wins over A57. A check-oracle win is assigned to the feasible configuration with the lowest check loss when it improves on A57; a validation-selected positive check win is assigned when validation selects an alternative configuration whose check loss improves on A57. These cases belong to the architecture-preselection pool and are disjoint from the margin-calibration and held-out target pools. The recorded S0 retained-indexed-configuration list contains three MLP and three GRU entries, including A57. No TCN configuration satisfies this rule. The PT-A57 reference candidate is constructed separately in the downstream candidate bank.

Two retained one-layer GRU-32 configurations have different indexed settings, but recurrent dropout is inactive for a one-layer GRU. They therefore represent the same executable architecture. The six retained configurations correspond to five executable architectures.

The final bank contains the pooled-source PT+FT initialization for the reference, one source-trained initialization for each of the five other retained indexed configurations, and a second A57

initialization used as a paired-initialization control. The six retained configurations therefore produce seven candidates. The bank and reference candidate are fixed before relative-margin calibration and held-out evaluation.

Source preparation is not identical across the candidate records. Strong-bank alternatives use 50 epochs of Adam/MSE training with learning rate 10^{-3} and batch size 64. The pooled-source PT-A57 reference is loaded from a separately frozen asset whose manifest records a nominal 50-epoch source recipe. The C1-A57 control has a distinct source-training lineage; its released source builder specifies 10 Adam/MSE epochs followed by 180 Adam/Huber refinement updates. The historical per-step C1 execution trace is not fully recoverable. Thus, the two A57 initializations differ in source preparation as well as initialization, and the controls do not isolate a random-seed-only or architecture-only effect.

S2.2 Relative-margin calibration records

Section 4.4 of the main manuscript defines the calibration rule. This section records its complete experimental instantiation and reproducibility details.

Let $\mathcal{T} = \{0.05, 0.075, 0.10, 0.125, 0.15, 0.20\}$ denote the preset margin grid. For a relative margin τ, let $\widehat{R}_{\mathrm{harm}}(\tau)$ denote the fraction of all calibration cases in which an alternative is selected and its check-set loss exceeds that of the adapted reference. Let $\widehat{G}_{\mathrm{ref}}(\tau)$ and $\widehat{G}_{\mathrm{pair}}(\tau)$ denote the mean relative check-loss gains with respect to the adapted reference and the paired-initialization control, respectively, where each gain is the relative reduction in check-set loss of the selected model with respect to the corresponding control. Their lower 95% confidence bounds are $\underline{G}_{\mathrm{ref}}(\tau)$ and $\underline{G}_{\mathrm{pair}}(\tau)$.

The eligible margins form

$$\begin{aligned}\mathcal{T}_{\mathrm{elig}} = \big\{\tau \in \mathcal{T} \;\big|\; & \widehat{R}_{\mathrm{harm}}(\tau) \le \rho, \\ & \widehat{G}_{\mathrm{ref}}(\tau) \ge \gamma_{\mathrm{ref}}, \quad \underline{G}_{\mathrm{ref}}(\tau) > 0, \\ & \widehat{G}_{\mathrm{pair}}(\tau) \ge \gamma_{\mathrm{pair}}, \quad \underline{G}_{\mathrm{pair}}(\tau) > 0\big\}.\end{aligned} \tag{S1}$$

The calibration uses $\rho = 0.05$ and $\gamma_{\mathrm{ref}} = \gamma_{\mathrm{pair}} = 0.03$. When $\mathcal{T}_{\mathrm{elig}}$ is nonempty, the calibrated margin is

$$\tau^* = \min \mathcal{T}_{\mathrm{elig}}. \tag{S2}$$

The smallest eligible value in the main margin-calibration setting is $\tau^* = 0.10$. The paired-initialization control is a two-candidate selector, not the unconditionally deployed second A57 model: both A57 source initializations are adapted on the same support set, then the reference-relative validation rule at the specified τ selects one for check evaluation. The $\widehat{G}_{\mathrm{pair}}$ denominator is the check loss of this paired-selector output in each corresponding calibration case.

Architecture preselection and candidate-bank construction are completed before margin calibration, and the resulting candidate bank and reference candidate remain fixed while the margin grid is evaluated.

If no margin satisfies the calibration criteria in a new domain, the adapted reference is retained. A margin fixed in another domain may still be evaluated without recalibration to assess transfer. The six-margin calibration results are given in Table S1. The figures are from the frozen 20-center, 80-case check-set calibration records; each center contributes four H/K cases. Cluster-resampled 95% confidence intervals use 4,000 repetitions. This is an empirical eligibility check, not a distribution-free risk guarantee.

Table S1: Five-criterion margin calibration on the main synthetic development pool (20 centers, 80 cases).

τ	Harm (%)	Ref. gain (%)	Ref. LCB (%)	Pair gain (%)	Pair LCB (%)	Eligible
0.050	6.25	13.93	7.75	13.33	7.73	No
0.075	6.25	13.79	7.77	13.18	7.39	No
0.100	5.00	13.85	8.06	13.32	7.68	Yes
0.125	5.00	13.13	7.01	12.60	6.85	Yes
0.150	5.00	12.70	6.48	12.17	6.30	Yes
0.200	2.50	10.98	5.56	10.45	5.70	Yes

Harm is the fraction of all calibration cases with a harmful alternative selected on the independent check set. Ref. and pair gains are mean case-wise relative check-MSE reductions against the adapted PT reference and the two-A57 paired-initialization selector, respectively; LCB denotes the lower bound of the corresponding 95% center-cluster bootstrap CI. Eligibility requires Harm $\leq 5\%$, both mean gains $\geq 3\%$, and both lower bounds > 0. The two smaller margins fail only the Harm criterion; $\tau = 0.10$ is the smallest eligible value. The exact frozen statistics are provided in the repository's C3.2 selector calibration manifest.

S3 Additional Analytical Details

S3.1 Ideal Candidate Decision within the Adapted Feasible Set

Let the expected future loss of adapted feasible candidate q on target center c be

$$R_c^{\mathrm{cand}}(q) = R_c\left(a_q, \theta_{c,q}^{(T)}\right). \tag{S3}$$

If this quantity were available, the ideal candidate-level decision would satisfy

$$q_c^{\dagger} \in \arg\min_{q \in \mathcal{Q}_c^{\mathrm{fea}}} R_c^{\mathrm{cand}}(q). \tag{S4}$$

This candidate-level quantity is unavailable during instantiation. MSA-DTI therefore uses post-adaptation validation loss $J_c(q)$ together with the relative-margin rule in the main manuscript.

S3.2 Conditional expected-loss result

Assume that validation loss differs from expected future loss by at most $\epsilon_c \geq 0$ for every feasible candidate:

$$\left|R_c^{\mathrm{cand}}(q) - J_c(q)\right| \leq \epsilon_c, \qquad \forall q \in \mathcal{Q}_c^{\mathrm{fea}}. \tag{S5}$$

When MSA-DTI selects an alternative, the relative-margin rule gives

$$R_c^{\mathrm{cand}}(q_c^*) - R_c^{\mathrm{cand}}(q^{\mathrm{ref}}) \leq -\tau J_c(q^{\mathrm{ref}}) + 2\epsilon_c. \tag{S6}$$

Hence, the selected alternative has lower expected future loss than the reference when

$$\tau J_c(q^{\mathrm{ref}}) > 2\epsilon_c. \tag{S7}$$

Equation (S7) is a sufficient condition under assumption (S5). The present experiments do not estimate an empirical bound on ϵ_c.

S4 Additional Experimental and Diagnostic Results

S4.1 Target-side baseline settings

For the synthetic main study, the window counts in support, validation, check, and test order are $(10, 1351, 628, 627)$, $(20, 1346, 625, 625)$, $(10, 1346, 624, 624)$, and $(20, 1342, 621, 621)$ for $(H, K) = (1, 10)$, $(1, 20)$, $(4, 10)$, and $(4, 20)$, respectively.

Table S2 reports the target-side settings used in the main comparison.

Table S2: Target-side settings of MSA-DTI and the baselines in the main comparison.

Method	Structure	Source initialization	Target update	Adapted models	Deployment-limit check
PT+FT	Fixed reference	Pooled-source model	SGD/MSE–50	1	Before output
MeDeT-based adaptation	Fixed reference	Meta-learned initialization	SGD/MSE–50	1	Before output
Random initialization	Fixed reference	Random	SGD/MSE–50	1	Before output
Few-shot NAS	Top-12 shortlist	Method-specific models	Adam/Huber–50	≤ 12	Before evaluation
Zero-shot NAS	Top-ranked model	Method-specific model	None	0	Before evaluation
Zero-shot NAS + fine-tuning	Top-12 shortlist	Method-specific models	Adam/Huber–50	≤ 12	Before evaluation
H-Meta-NAS-based adaptation	Target architecture search	Architecture-indexed meta-initializations	SGD/MSE–50	12.00	Before search evaluation
MSA-DTI	Candidate bank	Source-trained initializations	SGD/MSE–50	4–7	Before adaptation and output

PT+FT and MSA-DTI use the same source-trained reference and target adaptation procedure. Few-shot NAS and zero-shot NAS with fine-tuning use Adam/Huber ($\delta = 0.2$), learning rate 3×10^{-4}, weight decay 0, and a gradient-norm limit of 5.0. H-Meta-NAS-based adaptation uses 50 SGD/MSE updates per adapted model in the main comparison. It uses the same 66 architecture definitions, filters infeasible architectures before search, uses source-only first-order meta-initializations, and selects the final architecture using target validation data. Its settings are fixed before held-out evaluation. All methods use the same target cases, deployment limits, and architecture definitions.

The NAS labels in the figures and tables are abbreviated names of task-matched controls, not exact reproductions of published NAS algorithms. The *few-shot NAS* control combines source-prior ranking with a support-gradient entropy proxy, adapts up to 12 feasible models, and selects by validation MSE. *Zero-shot NAS* chooses one model with the support-gradient proxy but performs no target parameter updates; here "zero-shot" means no adaptation updates, not no access to labeled support. The fine-tuning variant adapts up to 12 proxy-ranked candidates before validation selection. The H-Meta-NAS-based control instead uses separate source meta-initializations.

S4.2 Paired held-out baseline comparisons

Table S3 summarizes the paired MSE comparisons between MSA-DTI and the main baselines on the 80 held-out cases.

Table S3: Paired MSE comparisons between MSA-DTI and the main baselines on the held-out cases.

Baseline	Strict MSE wins (%)	Mean paired MSE reduction (%)	95% CI (%)
PT+FT	71.25	14.60	[9.41, 20.14]
MeDeT-based adaptation	87.50	35.37	[25.39, 44.26]
Random initialization	100.00	80.71	[78.07, 83.22]
Few-shot NAS	95.00	40.40	[32.65, 47.58]
Zero-shot NAS	100.00	76.29	[68.95, 82.65]
Zero-shot NAS + fine-tuning	98.75	45.03	[36.53, 52.55]
H-Meta-NAS-based adaptation	100.00	68.94	[64.06, 73.53]

MSE win rate compares saved test MSE with the separately evaluated baseline using a strict inequality; ties are non-wins. For PT+FT, 13 of the 33 reference-retained cases count as strict wins because of numerical differences of at most 7.7×10^{-9} in test MSE, not additional beneficial selections. The Figure 6 alternative labels use the released 10^{-6} relative-gain cutoff, which agrees with strict test-MSE ordering for all 47 selected alternatives. Positive reductions favor MSA-DTI. The 95% intervals are the 2.5th and 97.5th percentiles of 4000 center-cluster bootstrap replicates, keeping each sampled target center's four cases together.

S4.3 Optimizer-matched diagnostic comparison

This comparison examines whether the performance ordering changes when target adaptation uses SGD/MSE and 50 parameter updates per adapted model. All methods use the same independent diagnostic target cases and deployment limits. The two common-shortlist controls use a shared 12-candidate set containing A57 and the 11 highest-ranked feasible alternatives under a fixed fused ranking (0.60 source-prior rank + 0.40 proxy rank). They differ only in the final selection rule: lowest validation loss versus the 10% reference-based margin. All selections are fixed before test evaluation.

Table S4: Full optimizer-matched comparison on the diagnostic target pool.

Method	MSE↓	Worst-10%↓	CVaR90↓	Adapted models
MSA-DTI	**0.003809**	**0.015094**	**0.008696**	6.10
PT+FT	0.004497	0.017144	0.011878	1.00
Few-shot NAS	0.006715	0.025246	0.019653	12.00
Zero-shot NAS + fine-tuning	0.007402	0.028099	0.020402	12.00
Common shortlist, lowest validation loss	0.006707	0.025240	0.019653	12.00
Common shortlist, reference-based selection	0.006819	0.025562	0.020547	12.00

MSA-DTI gives the lowest MSE, Worst-10% error, and CVaR90 in this comparison. Source initializations, selection rules, and numbers of adapted models remain method-specific, so the common per-model budget does not imply equal total target-side work.

S4.4 Selection-outcome details

On the 80 independent diagnostic cases, the relative-margin rule selects alternatives in 61 cases, including 49 beneficial and 12 harmful selections. Direct lowest-validation-loss selection chooses alternatives in 69 cases, including 55 beneficial and 14 harmful selections. The corresponding all-case harmful-selection rates are 15.00% and 17.50%, respectively.

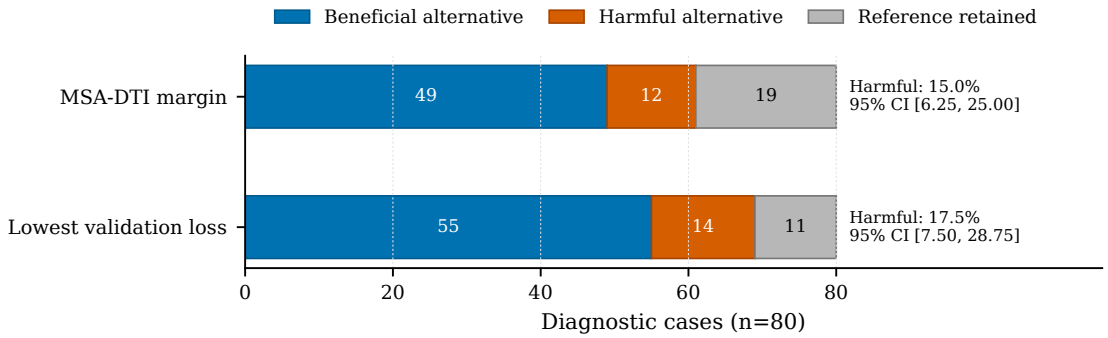


Figure S1: Selection outcomes on the 80-case diagnostic pool.

The relative-margin rule retains the reference more often. The individual harmful-selection-rate intervals overlap, but their overlap does not test the paired difference. The observed rate difference is reported descriptively.

S5 Timing, Hardware Profiling, and Deployment Analysis

S5.1 Target-side timing protocol

All methods are implemented in Python and PyTorch and run on an NVIDIA RTX 3060 Laptop GPU. Target-side time includes model creation, initialization loading, target-side search when applicable, target adaptation, validation evaluation, and final selection. It excludes offline preparation

and test evaluation. Each reported time is the mean over five runs, with the corresponding standard deviation reported in the main manuscript.

S5.2 Architecture-profiling protocol

The five executable architectures are profiled with batch size one on an Intel Core i7-12700H CPU and an NVIDIA RTX 3060 Laptop GPU for $H \in \{1, 4\}$. Each setting uses 100 warm-up inferences followed by five repetitions of 1000 timed inferences. CUDA execution is synchronized before and after each timed region. Median latency is the primary latency measure, and the 95th percentile summarizes upper-tail latency. Serialized parameter-file size is used as the measured model size. The increment in peak allocated CUDA tensor memory is recorded for each architecture–horizon pair.

Table S5: Per-architecture batch-one hardware profile.

Configuration	H	Operations	Parameters	CPU (ms)	GPU (ms)	State (KiB)	GPU Δpeak (KiB)
A1	1	155,712	77,921	0.097	0.148	306.5	1024
A1	4	155,904	78,020	0.094	0.133	306.9	1024
A6	1	157,760	78,977	0.093	0.168	311.0	1024
A6	4	157,952	79,076	0.099	0.151	311.4	1024
A13	1	159,808	80,033	0.128	0.237	315.5	1024
A13	4	160,000	80,132	0.128	0.218	315.9	1024
A55	1	377,888	2,081	1.793	3.999	10.3	54
A55	4	377,984	2,132	1.793	4.004	10.4	1030
A56/A57	1	1,050,688	5,697	1.980	4.019	24.4	60
A56/A57	4	1,050,880	5,796	1.975	4.170	24.8	1037

CPU and GPU values are median inference latency. A56 and A57 are the same executable one-layer GRU-32 architecture. GPU Δpeak is the nonnegative difference between the PyTorch CUDA maximum allocated tensor memory during the timed repetitions and the allocated-memory baseline recorded after warm-up and a cache clear, with peak statistics reset before timing. It excludes resident model/input allocations and is neither total device usage nor a configured deployment memory limit.

S5.3 Profiling results

For each device and $H \in \{1, 4\}$, the architecture-level correlation uses the five profiles displayed in Table S5, with one representative entry for the executable A56/A57 architecture. Across these four device–horizon groups ($n = 5$ distinct architectures per group), the Spearman correlation between estimated operation count and median inference latency ranges from 0.900 to 1.000. The correlation between parameter count and serialized parameter-file size is 1.000 in every measured setting, supporting parameter count as a model-size indicator. Peak GPU allocation is not monotonic in parameter count across the two horizons, so parameter count is not a direct proxy for runtime GPU memory. The MLP architectures use fewer estimated operations but more parameters than the GRU architectures. The two deployment indicators therefore capture different model properties, while their activity depends on the candidate set and limit tier. The profiling results characterize only the measured CPU/GPU host.

S5.4 Single-limit feasible-space analysis

To identify the active constraint in the main candidate bank, we apply the operation-count and parameter-count limits separately to the same seven candidates using their fixed complexity profiles.

For both $H = 1$ and $H = 4$, all seven candidates satisfy the operation-count limit under the tight, medium, and loose settings. Under the tight parameter-count limit, the three MLP candidates are excluded and the feasible bank is reduced from seven candidates to four, whereas the medium and loose parameter-count limits retain all seven candidates. Consequently, the joint feasible set

contains four candidates under the tight setting and seven under the medium and loose settings. The fixed reference candidate satisfies both limits in all six horizon–tier combinations.

S5.5 Expanded-bank dual-limit stress analysis

The preceding analysis shows that the operation-count limit is inactive for the seven-candidate main bank under the main limit tiers. To examine whether operation count and parameter count can independently restrict candidate feasibility, we additionally construct an eight-candidate stress bank by adding configuration 24 from the predefined architecture space.

The stress thresholds are derived solely from deterministic model-complexity profiles. Across $H \in \{1, 4\}$, the largest operation count in the original seven-candidate bank is 1,050,880, whereas the smallest operation count of configuration 24 is 1,198,112. Their midpoint gives the stress operation-count limit $B^F = 1{,}124{,}496$. Similarly, the largest parameter count among the stress-bank candidates other than A13 is 79,076, whereas the smallest parameter count of A13 is 80,033. Their midpoint gives the stress parameter-count limit $B^P = 79{,}554.5$.

For both $H = 1$ and $H = 4$, configuration 24 violates only the operation-count limit, A13 violates only the parameter-count limit, and the fixed reference A57 satisfies both limits. Each single-limit feasible set therefore retains seven of the eight candidates, whereas their joint feasible set retains six. Thus, the joint feasible set is a strict subset of each single-limit feasible set.

The target-side stress evaluation uses the same 80 held-out cases, and all models selected under the joint limits satisfy both limits. The expanded bank is used only for this limit-complementarity diagnostic, whereas the main evaluation uses the original seven-candidate bank.

S5.6 Selected-architecture analysis

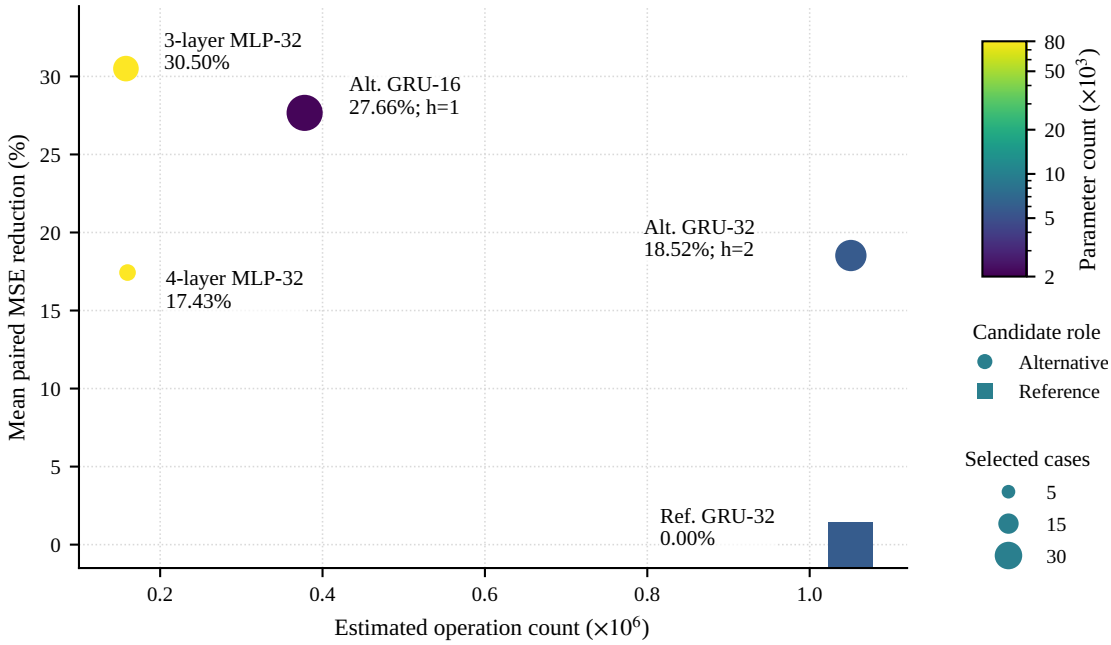
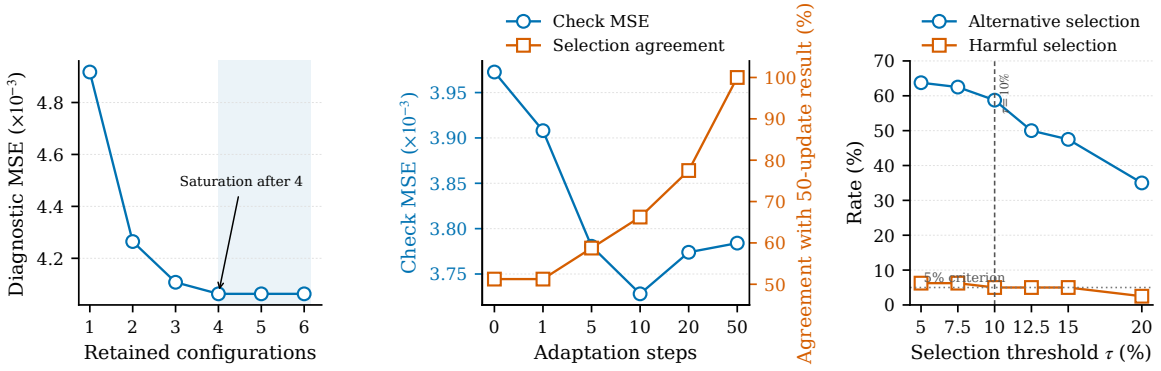


Figure S2: Selected-configuration complexity and performance. Marker area shows the number of selected cases, and h denotes harmful selections.

The reference configuration is retained in 33 cases. Four alternative configurations cover the remaining 47 cases, and their mean paired MSE reductions relative to the independently evaluated PT+FT baseline range from 17.43% to 30.50%. All four alternatives have positive mean reductions. The three harmful selections occur on two alternative GRU configurations.

S6 Configuration Analysis



(a) Retained configuration count (b) Target update budget (c) Threshold calibration

Figure S3: Configuration analysis of (a) the retained configuration count, (b) the target update budget, and (c) the relative margin.

Diagnostic MSE decreases from 0.004917 with one retained configuration to 0.004063 with four retained configurations and then remains unchanged. The main evaluation uses the six-configuration list fixed before evaluation. The observed saturation after four configurations suggests diminishing returns from adding retained configurations in this diagnostic; it does not identify an optimal bank size.

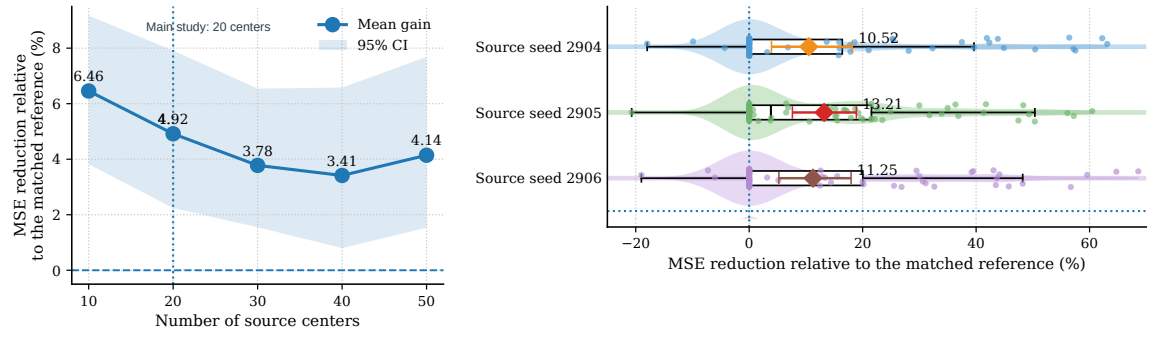
Most of the observed reduction in diagnostic check MSE occurs within the first ten target updates. The main experiment nevertheless uses 50 updates for every feasible candidate to preserve a common fixed target budget.

Using the same candidate bank, source initializations, feasibility rule, and reference-based selection rule at both endpoints, selected check MSE is 0.003973 with zero target updates and 0.003784 after 50 updates. The relative reduction computed from the two aggregate mean check-MSE values is 4.75%. This is a ratio of means, not the mean of paired case-wise relative reductions. The same candidate is selected at both endpoints in 51.25% of cases. An earlier interval for a different paired estimand cannot be assigned to this aggregate ratio without its original calculation record; accordingly, no interval-based benefit claim is made for this diagnostic.

The relative-margin analysis shows that 10% is the smallest value satisfying all preset calibration criteria.

S7 Source and Preselection Sensitivity and Real Workloads

S7.1 Source-center scale and source-training seeds



(a) Source-center scale (b) Source-training seeds

Figure S4: Synthetic sensitivity to (a) source-center scale and (b) source-training seeds.

The mean paired MSE reductions relative to the matched reference at source-center scales 10, 20, 30, 40, and 50 are 6.46%, 4.92%, 3.78%, 3.41%, and 4.14%, respectively. The mean gain remains positive at every tested source scale.

Across source-training seeds 2904, 2905, and 2906, the mean paired MSE reductions are 10.52%, 13.21%, and 11.25%. All three confidence intervals remain above zero, although some individual cases have zero or negative gains. The retained architecture list is held fixed in this seed study. The result therefore measures sensitivity to source-model training and initialization.

S7.2 Independent architecture-preselection stability

We separately repeat the complete 66-configuration diagnostic preselection on two new 20-center pools. Together with the original 20-center preselection pool, this gives three independently sampled preselection sets. The A57 reference, architecture space, initialization recipe, SGD/MSE–50 target recipe, and the preselection and retention rules defined in Supplementary Section S2.1 are kept unchanged. None of these pools overlaps the 20-center robustness pool used below, and no test data are used during preselection.

Table S6: Independent architecture-preselection stability and evaluation on centers 1160–1179.

Preselection pool	Centers	Retained configurations	Size	Jaccard	Gain (%)	95% CI (%)	Harmful (%)
Main S0	900–919	A1, A6, A13, A55, A56, A57	6	1.000	0.295	$[-0.646, 1.564]$	2.50
Independent S1	1120–1139	reference fallback only	0	0.000	0.000	$[0.000, 0.000]$	0.00
Independent S2	1140–1159	A55, A56, A57, A59	4	0.429	0.125	$[-0.894, 1.427]$	3.75

Jaccard similarity and Size are computed on the indexed-configuration lists retained by preselection, including A57 whenever it occurs in a list. The PT-A57 reference candidate is added separately to every downstream candidate bank; a retained A57 additionally supplies the paired C1-A57 candidate. Gain is the mean paired check-MSE reduction relative to PT-A57 on 80 robustness cases; intervals use a 4000-repetition center-cluster bootstrap. The diagnostic evaluates preselection stability while keeping the downstream procedure unchanged.

Across S0–S2, retained-list sizes are 6, 0, and 4 (median 4, range 0–6). A55, A56, and A57 are retained in two of three pools; A1, A6, A13, and A59 are each retained once. On the robustness pool, the S0 list selects PT-A57 in 72 cases, A55 in three, A56 in three, and the paired-control A57 initialization in two. S1 falls back to PT-A57 in all 80 cases. S2 selects PT-A57 in 71 cases, A55 in three, A56 in three, the paired-control A57 initialization in two, and A59 in one.

The two nonempty retained lists, S0 and S2, both have 95% confidence intervals for paired check-MSE gains relative to the adapted PT-A57 reference that include zero on the robustness pool, indicating sample sensitivity in candidate discovery. The main evaluation uses the S0 bank obtained from the main-study preselection pool, while the A57 reference provides a fallback when a preselection pool retains no alternative.

Reapplying the recorded preselection counts with retention thresholds of one, two, and three wins retains 10, 6, and 2 configurations, respectively. The threshold-two list used in the main study is S0, whereas threshold three retains only A55 and A57. On the existing step-50 diagnostic trajectory, threshold three increases selected check MSE by 3.79% relative to threshold two, with a 95% confidence interval of $[1.58\%, 6.30\%]$. The threshold-one list additionally contains A4, A5, A12, and A59. The main study did not generate source-trained records for these four additional configurations; downstream threshold sensitivity is therefore reported only for thresholds two and three.

S7.3 End-to-end propagation of preselection outcomes

To characterize the downstream effect of candidate-discovery variability, the recorded S0–S2 preselection outcomes are treated as fixed inputs and carried through the complete downstream procedure. Each outcome uses the same source centers and source-training recipe, followed by bank-specific application of the same margin-calibration rule on the same disjoint 20-center calibration

pool. The resulting pipelines are evaluated on a new 20-center held-out pool (centers 1180–1199; 80 cases). The architecture space, A57 reference, deployment limits, target adaptation protocol, margin grid, and calibration criteria remain unchanged. Held-out test data are not used in source training, calibration, deployment filtering, adaptation, or selection. For S2, A59 is source-trained using the same source-training recipe; S1 contains only the PT-A57 reference candidate.

Table S7: End-to-end propagation of the three preselection outcomes on centers 1180–1199.

(a) Candidate banks and calibrated margins

Preselection outcome	Retained configurations	Bank size	τ
S0	A1, A6, A13, A55, A56, A57	7	0.075
S1	Reference fallback only	1	–
S2	A55, A56, A57, A59	5	0.125

(b) Held-out performance and selection outcomes

Preselection outcome	Gain (%)	95% CI (%)	MSE win (%)	Alt. selected	Harmful (%)	Limit satisfaction (%)
S0	10.034	[4.602, 16.019]	52.50	48	7.50 / 12.50	100.00
S1	0.000	[0.000, 0.000]	0.00	0	0.00 / –	100.00
S2	8.784	[3.833, 14.238]	41.25	39	7.50 / 15.38	100.00

Gain is the mean paired test-MSE reduction relative to PT+FT. The 95% intervals use 4000 center-cluster bootstrap repetitions with the four cases of each target center kept together. Harmful rates are reported as all-case / conditional-on-alternative-selection percentages. For S1, no alternative is selected, so the conditional rate is undefined and shown as "–". The adapted PT-A57 reference is used in all S1 cases.

The two nonempty banks yield positive mean paired MSE reductions with confidence intervals above zero despite their different retained configurations. The S1 outcome contains no alternative and therefore follows the adapted-reference fallback without changing performance relative to PT+FT. These end-to-end results characterize the downstream consequence of the observed preselection variability; the retained-bank composition itself remains sample-sensitive.

S7.4 Causally corrected Alibaba workload evaluation

We use Alibaba Cluster Trace v2018 in a reconstructed causal-correction evaluation. The 20 source machines are deterministically reconstructed, and the 20 calibration and 40 held-out machine identities match the retained historical cohorts; their roles remain disjoint. The corrected source models are retrained on an independent source pool. A source-derived 10-second event- time grid supplies the fixed prediction opportunities, with event time assumed to be the observation-availability time. The input length is $L = 96$ and the horizon is $H \in \{1, 4\}$. The response is the next H genuinely observed CPU values scaled from 0–100 to fractions in $[0, 1]$, so MSE has units of CPU-fraction squared. The 25-channel input uses historical values, forward-only missing-value handling, trailing-six CPU statistics, and causal observation masks. Source MKPI values were unavailable in the registered source interval; its value and mask channels are identically zero for all methods. This correction therefore does not establish performance with MKPI.

For each target case, adaptation uses $K \in \{10, 20\}$ valid labeled support windows, followed by 80 distinct validation windows; the calibration cases add 60 check windows. Held-out testing evaluates every label-valid origin within 320 fixed candidate opportunities (290–319 windows per case). Missing labels are not interpolated, and every method uses the same eligible evaluation origins. Deployment-limit tiers remain controlled settings rather than machine-measured resource limits. The main and paired selectors are based on separately source-trained initializations, with the paired control selecting between the two A57 candidates. Their source-training procedures are not identical, so this is not a seed-only control.

The full five-criterion rule of main Eq. (12) is applied to all six fixed margins on the 80 calibration cases, with a 20-machine, 4,000-repetition paired cluster bootstrap (seed 2904). The reference and

paired gains are case-wise relative reductions in check-set MSE. Table S8 reports the corresponding mean gains and 95% lower confidence bounds.

Table S8: Full five-criterion calibration for the causally corrected Alibaba cases (20 machines, 80 cases).

τ	Harm (%)	Ref. gain (%)	Ref. LCB (%)	Pair gain (%)	Pair LCB (%)	Eligible
0.050	15.00	-6.76	-15.06	-2.05	-8.51	No
0.075	15.00	-6.96	-15.29	-2.25	-8.63	No
0.100	8.75	-5.68	-13.70	-3.33	-10.18	No
0.125	8.75	-5.68	-13.70	-3.33	-10.18	No
0.150	6.25	-4.11	-11.04	-2.28	-8.41	No
0.200	2.50	-1.43	-6.98	-1.43	-6.98	No

Harm is the fraction of all calibration cases with a selected alternative whose check loss exceeds the reference. Eligibility requires harm $\leq 5\%$, both mean gains $\geq 3\%$, and both respective 95% lower bounds > 0. All six margins fail the complete rule.

Consequently, the calibrated MSA-DTI policy retains the adapted PT-A57 reference for all 160 held-out cases. It has the same mean MSE (0.00338831) as PT+FT and an identical-case paired improvement of 0 by construction; this is not evidence of a population-level harmful-selection guarantee. The separately pre-registered fixed-$\tau = 0.10$ diagnostic selects nine alternatives, including three harmful replacements (3/160, or 1.875% of all held-out cases). The mean of case-wise relative MSE gains against PT+FT is -0.2592% (machine-cluster 95% CI $[-1.4711\%, 0.6205\%]$). Although its mean MSE (0.00337210) is slightly below the PT+FT mean, that ratio of mean MSEs is not the reported mean paired relative gain. The diagnostic does not establish positive external-domain transfer.

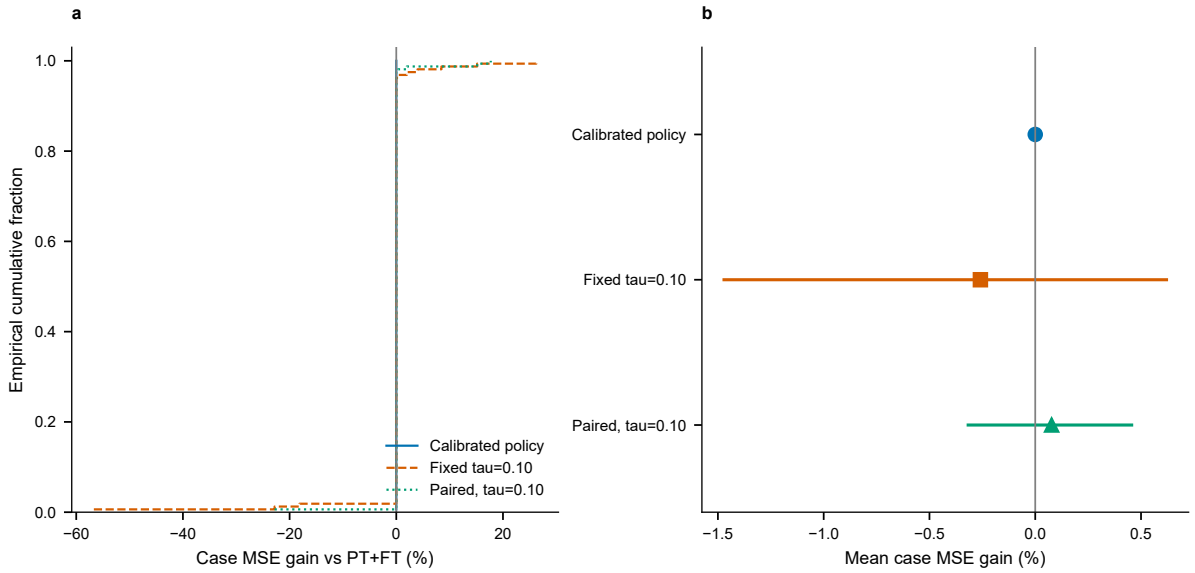


Figure S5: Causally corrected Alibaba held-out comparisons against PT+FT: (a) case-level relative-gain distributions and (b) machine-cluster 95% confidence intervals for mean case-wise relative gains. Fixed-margin results are separate diagnostics; the calibrated policy retains the reference.

The corrected evaluation uses a previously observed machine population; it is a post-observation correction, not an independent blind study. Arrival times were not recorded and event timestamps are assumed to be availability times. The earlier noncausal Alibaba result is preserved as historical evidence but is not used to support the present external-domain claims. The corrected reproducibility archive containing protocol locks, model checkpoints, case-level records, and plots is publicly available in the dated evidence-only Release cited in the main manuscript's Data Availability statement. The third-party raw trace and large corrected input/label arrays are not redistributed.

S7.5 Bitbrains in-domain VM workload evaluation

The publicly released GWA-T-12 Bitbrains fastStorage trace[1] contains native VM-level CPU, memory, storage, and network observations. A registered quality filter was applied to the full set of 1,250 VMs, after which 100 eligible VMs were selected deterministically by their stable identifiers. Disjoint sets of 20 source, 20 development, 20 calibration, and 40 held-out VMs were fixed before model evaluation. This is an in-domain model-instantiation experiment: source initializations are trained anew on Bitbrains source VMs rather than transferred as frozen synthetic or Alibaba weights. The prediction target is recorded CPU usage in percent divided by 100; values above 100% are retained as recorded, not clipped. The input uses source-only normalization, causal historical features, and role-disjoint chronological windows; $H = 1, 4$ means next one or four native records. Typical elapsed horizons are about 300 and 1200 seconds, although irregular record gaps cause variation.

Frozen Bitbrains data protocol.

Item	Definition or actual per-VM window counts
Qualification	Fixed 11-column schema; finite integer Unix-second timestamps in $[10^9, 2 \times 10^9)$, strictly increasing; finite nonnegative CPU labels; at least 3,000 post-first-day records and 50 complete source windows for each horizon. Every H/K case requires exactly K support windows and at least 80/60/200 validation/check/test windows. These predicates yield 1,082 eligible VMs before identity selection.
Timeline	Source uses records in $[t_0, t_0 + 86{,}400\,\text{s})$, where t_0 is each VM's first timestamp (observed first-to-last span 86,107 s); target uses the first 3,000 native records at/after that cutoff (observed spans 899,799–903,397 s). Support contains $96 + H + K - 1$ raw records; remaining records follow normalized 0.2:0.1:0.1 validation/check/test allocation, with floors then largest remainders and earlier-segment tie priority.
Channels	Ten raw fields: CPU cores, capacity, MHz usage and percent usage; memory capacity/usage; disk read/write and network received/transmitted throughput. Add trailing-five CPU mean and first difference, their 12 corresponding validity masks, and preceding elapsed seconds/1,200. Invalid inputs use causal forward fill from initial zero; masks retain invalidity. Fill, mean, difference and gap state restart at every raw role boundary.
$H = 1, K = 10$	(10, 1,351, 628, 627)
$H = 1, K = 20$	(20, 1,346, 625, 625)
$H = 4, K = 10$	(10, 1,346, 624, 624)
$H = 4, K = 20$	(20, 1,342, 621, 621)
Frozen entry	Public protocol asset: `code/experiments/r4_r5/bitbrains_data.py` and `bitbrains/data/bitbrains_dataset_manifest.json`.

Count order is support, validation, calibration check, and held-out test; counts are constant within the corresponding VM groups. Validation applies to calibration and held-out cases; development-fixed uses support/check only. These are windows, which may share label endpoints.

Within each held-out case, MSA-DTI and each comparator use the same observed Test targets. Support adaptation uses $K = 10$ or 20 windows per model with 50 SGD/MSE updates. The pretrained seven-token bank, per-model feasibility checks, and the five relative-margin criteria remain as in the synthetic method; calibration is performed on distinct VMs. For the development-fixed control, one feasible candidate is selected in advance for each budget tier and horizon using average development check MSE. It alone is adapted at the target; it does not inspect target validation for architecture selection. The full MSA-DTI selector instead adapts the 4 or 7 feasible candidates before comparing their post-adaptation validation scores. Thus, the method and fixed control share the candidate bank and target protocol but not the total number of candidate adaptations.

[1] Dataset: https://atlarge-research.com/gwa-t-12/. The trace was provided by Bitbrains IT Services Inc.

Table S9: Full five-criterion relative-margin calibration on Bitbrains development-disjoint calibration VMs (20 VMs, 80 cases).

τ	Harm/all cases	Gain vs. reference (95% CI)	Gain vs. paired selector (95% CI)	Eligible
0.050	0.00%	38.56% [25.64%,51.83%]	19.36% [10.30%,29.46%]	Yes
0.075	0.00%	38.39% [25.37%,51.65%]	19.39% [10.26%,29.47%]	Yes
0.100	0.00%	38.01% [24.87%,51.40%]	19.01% [9.96%,29.03%]	Yes
0.125	0.00%	38.01% [24.87%,51.40%]	18.98% [9.95%,29.01%]	Yes
0.150	0.00%	37.81% [24.59%,51.31%]	18.97% [9.95%,28.98%]	Yes
0.200	0.00%	37.81% [24.59%,51.31%]	18.97% [9.95%,28.98%]	Yes

All rows pass all five criteria, including the preset 5% all-case harmful rate, both minimum 3% gains, and both strictly positive 95% lower bounds. The least restrictive qualifying margin, $\tau = 0.05$, is frozen before held-out evaluation. Intervals use 4,000 VM-cluster bootstrap repetitions. These empirical calibration conditions are not a held-out or population guarantee.

For the 160 held-out cases, MSA-DTI achieves a 32.7045% mean of case-wise relative test-MSE improvements over PT+FT (95% VM-cluster CI [7.6721%,51.2506%]); 123 cases improve, 11 worsen, and 26 are equal. The mean case MSE is 0.00742182 for MSA-DTI versus 0.00807491 for PT+FT. Their ratio of aggregate mean losses corresponds to an 8.0879% reduction; this secondary estimand is not interchangeable with the mean of case ratios. The comparison against the development-fixed control yields a descriptive 38.3851% mean case-relative gain (95% VM-cluster CI [25.5620%,50.8000%]). The held-out harmful-selection frequency is 11/160 (6.8750%), exceeding the 5% empirical calibration criterion. The primary evidence supports improvement in this VM cohort but not universal risk control or benefit on all real traces, as the corrected Alibaba result demonstrates.

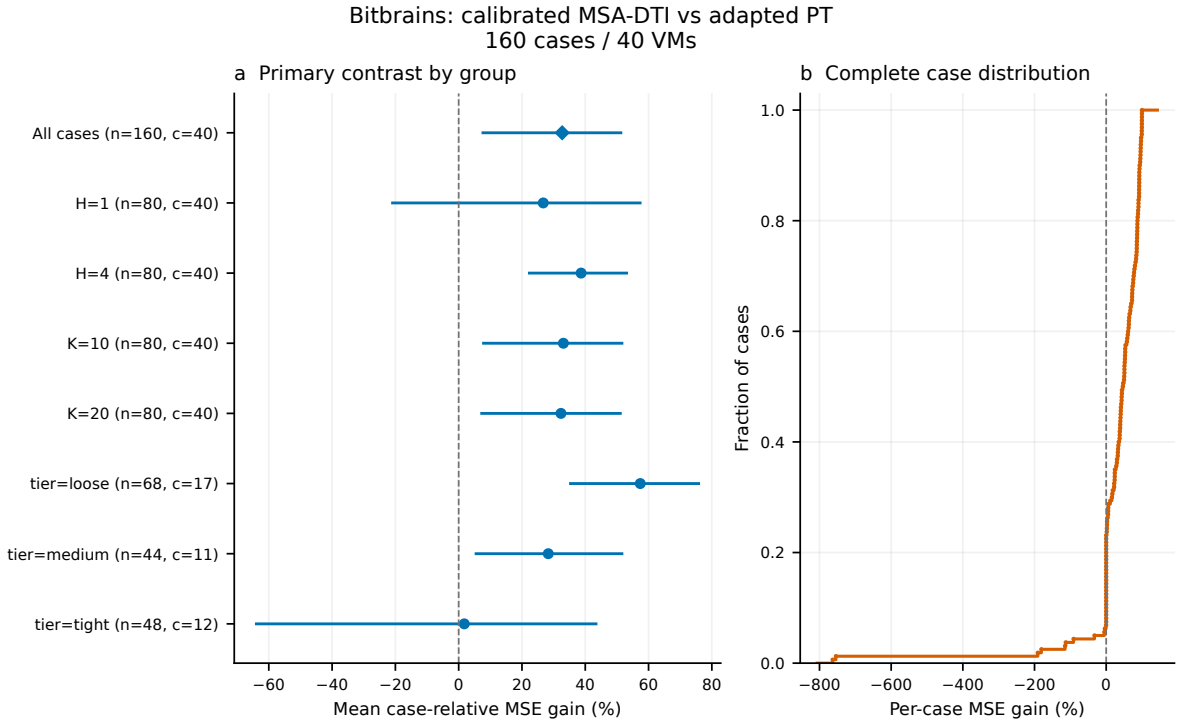


Figure S6: Bitbrains fastStorage case-relative held-out MSE gains for calibrated MSA-DTI versus PT+FT. Intervals use 4,000 paired VM-cluster bootstrap draws; subgroup comparisons are descriptive without multiplicity correction. All registered cases, including large adverse relative values, are retained.

The mean target-side ready-tensor processing time is 19.4735 seconds per MSA-DTI case versus 4.7403 seconds for PT+FT, with mean numbers of adapted models 6.10 versus 1.00. Shared

execution components limit strict interpretation as independent end-to-end timing runs. Hidden VM host/tenant structure, a single source-training realization, and assumed event-time data availability also limit broader generalization. The primary fastStorage cohort was selected before model outcomes; the optional Rnd trace was not run in this study and is not an unreported negative or positive result.

S7.6 Strong development-fixed candidate comparison on synthetic centers

A targeted supplementary contrast evaluates whether target-specific post-adaptation selection improves on committing in advance to a strong candidate from the same seven-token source-trained bank. For each budget and horizon, an independent development-center pool selects the lowest mean-check-MSE feasible token after averaging the two support sizes per center; the chosen token is frozen before the 80 evaluation cases. During evaluation, the fixed comparator adapts only that token for 50 updates, whereas MSA-DTI adapts all feasible tokens under the same per-model update rule and uses the fixed synthetic margin $\tau = 0.10$ for validation selection. The fixed comparator therefore has lower total target effort.

The MSA-DTI mean case-relative test-MSE improvement over this *Dev-Fixed-Best* comparator is 13.4803% (95% center-cluster CI [6.3856%,21.0963%]); 43 cases improve, 12 worsen, and 25 are equal. MSA-DTI requires 5.95 adapted models and 297.5 updates per case on average, versus 1 model and 50 updates for Dev-Fixed-Best. Mean ready-tensor target-side time is 15.5813 versus 1.8262 seconds, respectively. *Evidence-status limitation:* the new development and evaluation pools do not overlap any reconstructable historical center/seed domains, but the original RNG identities of two historical R2 screening pools are not recoverable. This comparison is retained as post-audit supplementary support, not as strictly confirmed independent evidence closing every historical development-pool separation requirement.

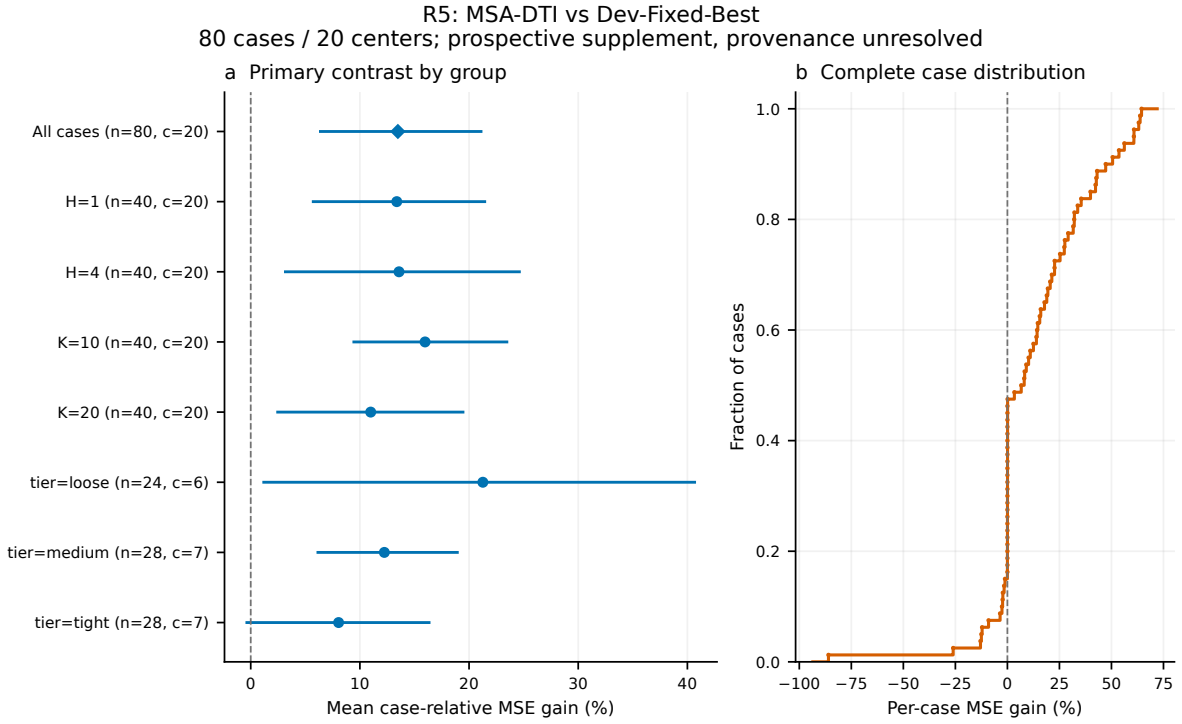


Figure S7: Additional synthetic same-bank strong-fixed control: case-relative test-MSE gains of MSA-DTI over Dev-Fixed-Best. Paired intervals resample 20 centers; historical development-pool RNG provenance is not fully closed.

S8 Synthetic Benchmark Generation and Center Heterogeneity

The controlled benchmark represents heterogeneous computing centers in a common metric coordinate space. Each center may observe only a subset of the metric positions. The model input uses 12 value channels, 12 corresponding observation-mask channels, and one time-gap channel, giving a fixed input dimension of 25. Unavailable metric values are zero-filled and identified by the corresponding mask channels.

Center types A–C are evaluation strata rather than model classes. They differ in workload dynamics, sampling, and monitoring conditions. Deployment-limit tiers are separate platform-side settings and are assigned using the type-dependent probabilities reported in Table S10. The table also gives the complete type-specific parameter ranges. The default A:B:C ratio is 3:4:3, giving 6 Type-A, 8 Type-B, and 6 Type-C centers in each 20-center source or held-out target pool.

Table S10: Type-specific parameter ranges of the controlled synthetic benchmark.

Parameter	Type A	Type B	Type C
Sampling interval d_t	$\{1\}$	$\{1, 2\}$	$\{2, 3, 4\}$
Point-missing rate	$[0.02, 0.08]$	$[0.10, 0.25]$	$[0.25, 0.45]$
Missing-block length	$[5, 15]$	$[20, 60]$	$[60, 120]$
Observation-noise std.	$[0.01, 0.03]$	$[0.03, 0.06]$	$[0.06, 0.10]$
Permanently unavailable metrics	0–2	2–4	4–6
Base workload logit	$[-0.70, 0.10]$	$[-0.45, 0.35]$	$[-0.20, 0.55]$
Seasonal amplitude	$[0.20, 0.45]$	$[0.30, 0.60]$	$[0.40, 0.75]$
AR coefficient	$[0.55, 0.80]$	$[0.60, 0.88]$	$[0.65, 0.92]$
Process-noise std.	$[0.04, 0.08]$	$[0.06, 0.11]$	$[0.08, 0.14]$
Burst rate per 1000 base steps	$[1.0, 2.5]$	$[1.5, 3.5]$	$[2.0, 4.5]$
Burst amplitude	$[0.25, 0.55]$	$[0.35, 0.75]$	$[0.45, 0.95]$
Drift probability	0.20	0.40	0.60
Drift strength	$[0.10, 0.25]$	$[0.15, 0.35]$	$[0.20, 0.45]$
Limit-tier probabilities (tight / medium / loose)	0.25/0.45/0.30	0.35/0.45/0.20	0.45/0.40/0.15

The three types share the same missingness mechanisms. For every center, the generator may create 2–6 single-metric outage blocks and 1–3 grouped outage blocks containing 2–4 metrics. A short global monitoring outage occurs with probability 0.20. The shared periodic base scales are 24, 96, and 672 base steps.

Together, these settings vary five aspects of the benchmark: workload dynamics, sampling, monitored-variable availability, observation conditions, and deployment-limit settings.

For each center, the generator first creates a latent workload process. The workload contains periodic variation, autoregressive variation, and random workload bursts. When drift occurs, it is generated as either an abrupt change or a gradual transition.

The latent workload is then mapped to multivariate monitoring signals. Individual metrics may have different delays, gains, offsets, nonlinear terms, and observation noise. Center-specific sampling is applied before monitoring-schema and missingness effects are introduced.

The benchmark generator and all data-role assignments are fixed before architecture preselection, margin calibration, diagnostics, and held-out evaluation.